



Modulatory Sedative Activity of Helicide on Diazepam in Thiopental Sodium Mediated Sleeping Mice: An *In Vivo* Approach With Molecular Docking

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Abstract

Insomnia is a common sleep disorder characterized by difficulties in falling asleep, staying asleep, or getting restful sleep. This study aimed to assess the sedative properties of helicide using thiopental sodium-induced sleep mice model. Helicide (5 and 10 mg/kg) and diazepam (2 mg/kg) were administered to the animals. After 30 min, thiopental sodium (20 mg/kg) was given (*i.p.*) to induce sleep, and onset and duration of their sleep were recorded over a 4-h period. Molecular docking was also performed to investigate the potential interactions of helicide with GABA_A (PDB: 6X3X, containing α 1 and β 2 subunits) receptors and assess its pharmacokinetics and toxicity. The results demonstrated that helicide significantly ($p > 0.05$) extended the duration of sleep (174.8 ± 10.47 min) and provided fast sleeping onset (9.40 ± 1.71 min) compared to control. The combination of helicide and diazepam elevated sedation duration (191.40 ± 3.86 min) and lowered onset (5.20 ± 0.58 min), indicating a synergistic effect. Helicide exerted binding affinity of -7.7 kcal/mol, while diazepam had a slightly higher affinity of -8.7 kcal/mol with 6X3X. Helicide also displayed favorable pharmacokinetics and drug-likeness. Overall, Helicide exhibited a moderate sedative effect in the mice model, likely through its interaction with the GABAergic system.

Keywords Sedation · Insomnia · Molecular docking · GABA agonist · Toxicity · Pharmacokinetics

Introduction

Insomnia is a major public health concern characterized by difficulty falling or staying asleep, early morning awakenings, non-restorative sleep, and daytime impairments (Agu et al. 2023). About 25% of people worldwide suffer from sleep disturbances at the moment. Individuals who experience sleep disturbances all year round are more vulnerable to mental health conditions like depression, bipolar disorder, generalized anxiety disorder, and suicidal thoughts. Depression-related disorders constitute one of the most often diagnosed mental conditions, with a lifetime frequency of around 16% (Ohayon 2002; Morin et al. 2006). According to the Centers for Disease Control and Prevention, around 14.5% of adults struggle with falling asleep, while 17.8% have trouble staying asleep. Insomnia prevalence varies by age, gender, and ethnicity, with women being more affected than men (<https://www.cdc.gov/nchs/data/databriefs/db436.pdf>, accessed on February 9, 2025). Insomnia arises from neurodegenerative, inflammatory, traumatic, or ischemic

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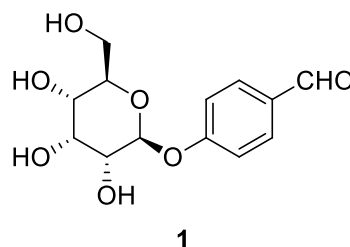
changes in sleep-regulating brain regions, affecting neurotransmitters. It is also linked to co-morbidities, neurological diseases, medications, and conditions like stroke, epilepsy, and dementia (Mayer et al. 2021).

Neurotransmitters such as gamma-aminobutyric acid GABA, norepinephrine, histamine, dopamine, glutamate, serotonin, epinephrine, and hypocretin significantly influence sleep duration, making them potential targets for insomnia treatment (Ferdous et al. 2024). Gamma-aminobutyric acid is a crucial neurotransmitter that facilitates sedation and is the predominant inhibitory neurotransmitter in the central nervous system (Brohan and Goudra 2017). Gamma-aminobutyric acid receptors consist of three subtypes: GABA_A, GABA_B, and GABA_C (Terunuma 2018). The activation of the GABA_A receptor facilitates the inward flow of chloride ions through the opening of the ligand-gated ion channel, resulting in hyperpolarization and the inhibition of signal transmission (Patil and Gupta 2016).

In Europe and various regions worldwide, traditional medicine, including traditional Chinese, traditional African, and Islamic medicine, is often employed for treating mental illnesses (Sarris et al. 2011; Kpobi et al. 2019; Qu et al. 2019; Mitha 2020). Evidence supports the effectiveness of traditional medicines in treating mental illnesses. For instance, hypericin, found in *Hypericum perforatum* L., Hypericaceae (St. John's wort), acts as a key pharmacologically active compound and is sold over-the-counter in the US as an herbal remedy for improving mood. Similarly, dietary supplements containing St. John's wort demonstrate antidepressant effects. Studies also indicate that Chinese polyherbal remedies such as decoctions of *gan-mai-da-zao* (Jun et al. 2014), *xiao-yao-san* (Yuan et al. 2020), and *chai-hu-shu-gan-san* (Sun et al. 2018) effectively alleviate symptoms associated with depression. Many herbal remedies and natural products have demonstrated antidepressant properties through multiple randomized clinical trials and animal studies (Yeung et al. 2018).

Helicide (1, 4-formylphenyl-β-D-allopyranoside), a key compound found in *Helicia nilagirica* Bedd., Proteaceae, is renowned for its analgesic and sedative-hypnotic properties, and it has also been employed in the treatment of psychoneurosis (Liu et al. 2005; Lan et al. 2007; Li et al. 2011). Research has shown that helicide and its derivative W0620 exhibit antidepressant properties in both the forced swimming test and the mouse tail suspension test (Rong 2007; Tong et al. 2008, 2019). Helicide reversed symptoms of depression, such as impaired movement, reduced preference for sucrose, lower emotional behavior scores, and weight loss, and it significantly increased the number of bromodeoxyuridine (BrdU)-positive cells in the dentate gyrus of the hippocampus in chronically stressed mice, along with enhancing the expression of brain-derived neurotrophic factor (Tong et al. 2010, 2012). Scientists are increasingly interested in helicide because of its well-documented

neurological effects and minimal side effects (Xia et al. 2019). This study aimed to investigate the sedative effect of helicide on thiopental sodium (TS)-induced sleep in mice. Furthermore, a combination study of helicide (1) and diazepam was conducted to assess their synergistic effect. Additionally, computer simulations were employed to examine helicide's interactions with the target receptor.



Materials and Methods

Chemical Reagents

Diazepam and water for injection were supplied by Square Pharmaceuticals and Opsonin Pharmaceuticals Ltd., Bangladesh, respectively. Helicide (1, 98% purity, batch number: S379701) was acquired from Sigma-Aldrich (USA), and Tween 80 was purchased from Merck (Mumbai, India).

Test and Control Groups

Following an extensive literature review, we determined two effective doses for the test compound (helicide) a higher dose (10 mg/kg) and a lower dose (5 mg/kg) (Tong et al. 2010, Tong 2012). First, we prepared a mother solution of helicide at a concentration of 10 mg/kg for the higher dose by dissolving the compound in water for injection. From this mother solution, dilutions were prepared to achieve a lower dose of 5 mg/kg. Next, a solution of diazepam in water for injection was prepared to reach a final dose of 2 mg/kg. Finally, for the combination therapy, co-administration of the higher dose of helicide with diazepam was administrated to explore their combined effects more thoroughly. This methodical approach ensured precise dosing and allowed for an accurate assessment of both the individual and combined effects of helicide and diazepam.

Experimental Animals

Swiss albino mice, weighing between 22–25 g, were obtained from the animal house of Jahangirnagar University. They were carefully housed in the pharmacology lab at Bangabandhu Sheikh Mujibur Rahman Science and Technology University, under controlled conditions. The environment

maintained a stable temperature of 28 ± 1 °C, with 65% relative humidity and a consistent 12-h light/dark cycle to regulate their natural rhythms. Throughout the study, the mice were provided with unrestricted access to standard food and water, ensuring their comfort and well-being. The study was carried out from 8 am to 3 pm, followed by a 17 h observation period to monitor for any potential post-study effects.

In vivo Protocol

Following a three-day acclimatization period, a total of 25 animals were randomly assigned to one of five experimental groups, each consisting of five animals: control (vehicle 10 ml/kg), diazepam (2 mg/kg), helicide (Hel-5) (5 mg/kg), Hel-10 (10 mg/kg), and Hel-10 + diazepam (10 + 2 mg/kg). Treatments were administered intraperitoneally, with groups receiving either the vehicle (control), the standard reference drug diazepam, or one of the test compounds. After a 30 min pre-treatment interval, thiopental sodium was administered at a dose of 20 mg/kg (*i.p.*) body weight to induce sleep. Each animal was then placed into an individual observation chamber, such as a plastic cage, for continuous monitoring. The primary outcome measures were the latency to sleep onset and the duration of sleep for 4 h. Sleep latency was defined as the time from thiopental sodium administration to the loss of the righting reflex, while sleep duration was measured as the time from the loss to the recovery of the righting reflex. Both parameters were recorded manually with careful observation to ensure accuracy. The treatments and their respective doses are detailed in Table 1.

Statistical Analysis

The sedative effectiveness results are presented as the mean and standard error of the mean (SEM). Statistical analysis was performed using Graph Pad Prism (version 6.0), a software program designed to determine the significance of observed fluctuations, with a confidence level of 95%. Statistical significance was defined as *p* values below 0.05.

Table 1 Different treatments including controls and tests, were provided to the experimental animals, and their administered doses in sedative test

Treatment Group	Dose (mg/kg)	Target Receptor
Control (Vehicle)	10 ml/kg	-
DZP	2	GABA _A
HEL-5	5	Under investigation
HEL-10	10	Under investigation
HEL-10 + DZP	10 + 2	Under investigation

Control (vehicle): Tween 80 and distilled water; DZP: Diazepam (Dose: 2 mg/kg); HEL: helicide (Dose: 5 and 10 mg/kg); All treatments provided through intraperitoneal (*i.p.*) route

In silico Analysis

Selection and Preparation of GABAA

The $\alpha 1$ and $\beta 2$ subunits of the GABA_A receptor, which have been identified in earlier studies as key mediators of sedation (Atack 2005), were selected for investigation in this study. The Protein Data Bank (PDB) is the primary global repository that offers open access to three-dimensional (3D) structures of biological macromolecules and their complexes, serving as a crucial resource for biological research (Burley et al. 2017). The 3D structures for the GABA_A receptor subunits, specifically $\alpha 1$ in the B chain and $\beta 2$ in the A chain (PDB ID: 6X3X), were obtained from the RCSB Protein Data Bank (<https://www.rcsb.org/>, accessed on Sept 8, 2024). To reduce potential interference during docking simulations, the receptors were optimized after their initial collection. This optimization was carried out using PyMOL software (version 1.7.4.5), which involved removing extraneous amino acid residues, water molecules, and heteroatoms from the macromolecules (Bhuia et al. 2023). Following this, the protein structures underwent energy minimization using the GROMOS96 43 B1 force field with Swiss-PDB Viewer software (Saleh et al. 2016).

Collection and Preparation of Ligands

The chemical structures of helicide (PubChem ID: 12896796) and diazepam (PubChem ID: 3016) were sourced from the PubChem database and were available in the ‘sdf’ file format. The Allinger MM2 force field was employed to reduce the energy of the ligands using Chem3D Pro 21.0 software (Son and Rybolt 2013).

Docking Protocol and Non-bond Interaction

Molecular docking simulation is a widely utilized computational technique in pharmaceutical research for drug development (Pinzi and Rastelli 2019). This approach evaluates the pharmacodynamic properties of drugs by examining the interactions between molecules and specific binding sites using the PyRx v0.8 software (Mukty et al. 2024). The results of the docking process reveal the interaction strength between a ligand and a protein’s active site. To accelerate the docking process, the dimensions of the grid box were modified with the X, Y, and Z axes set to $35.02 \times 39.37 \times 25.00$ Å. The computation was run 2000 times. The ligand–protein complex was first saved in PDB format to extract the ligand in PDBQT format (Hetényi and Bálint 2020). The binding affinity of the ligand was reported as a negative value in kcal/mol. Additionally, the active binding sites of the proteins were examined using BIOVIA Discovery Studio v21.1.0, which enables the assessment of non-bond interactions within ligand–protein complexes (Le and Do 2022).

Drug Likeness

Pharmacokinetics is the study of a drug's absorption, distribution, metabolism, and excretion (ADME), considering factors that affect its ability to reach its target in the body. Estimating the ADME properties early in the drug discovery process can significantly reduce the likelihood of pharmacokinetics-related failures in later clinical stages (Khatun et al. 2024; Hay et al. 2014). The SwissADME and pkCSM (<https://biosig.lab.uq.edu.au/pkcsm/prediction>) online server can be utilized to predict ADMET parameters and assess drug-likeness properties (Hasan et al. 2024).

Toxicity Prediction

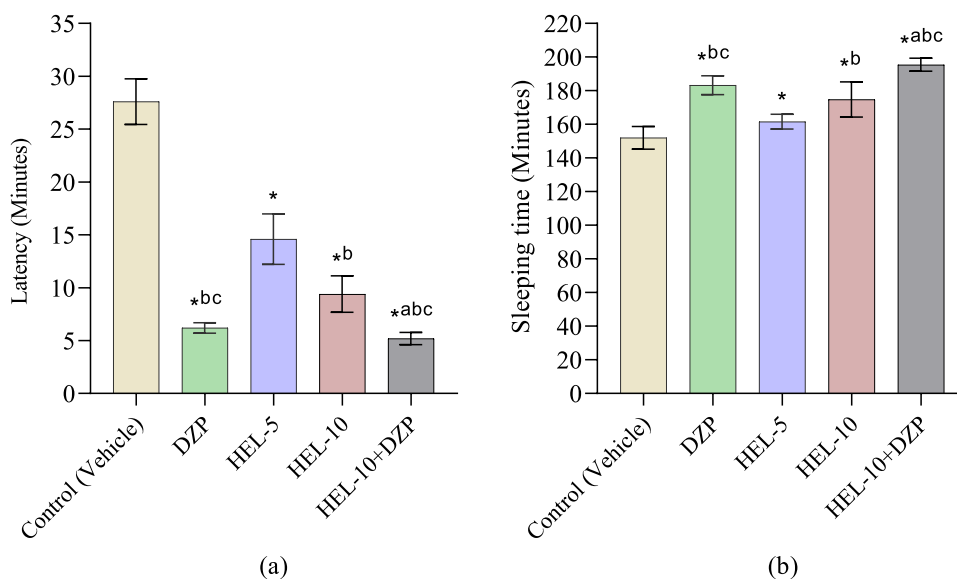
Toxicity prediction is a key part of drug discovery, helping to find and rank compounds that are more likely to be safe and effective in humans. It also lowers the chances of expensive failures later in drug development (Issa et al. 2017; Amorim et al. 2024). The ProTox 3.0 web server can be used to predict different toxicity parameters for a given drug. To evaluate these parameters, the canonical SMILES from PubChem were uploaded to the ProTox 3.0 platform (<http://tox.charite.de/protox-3>). The server then identified the toxicity class and other characteristics of the chemical. The toxicity parameters of the selected compounds were analyzed and recorded in Table S2.

Results and Discussion

In Vivo Investigation

The study findings are presented in Fig. 1, evaluating different treatment groups (from control to helicide + diazepam)

Fig. 1 *In vivo* investigation to evaluate different treatment groups from control to helicide (1) in conjunction with diazepam based on their sleeping latency and total sleeping time. **a** Latency, **b** sleeping time in min observed in different treatment groups of animals. Values are mean $\pm$ SEM ($n=5$); one-way ANOVA followed by *t*-students Newman-Keuls as a *post-hoc* test with multiple comparison; $p < 0.05$ compared to the *control (Vehicle). Abbreviations: ^adiazepam (DZP), ^bhelicide 5 mg/kg (HEL-5), ^chelicide 10 mg/kg (HEL-10), ^dHEL-10 + DZP; control: vehicle (distilled water containing 0.9% NaCl and 0.5% tween 80)



based on their sleeping latency and total sleeping time. The findings indicate that helicide administration significantly ($p < 0.05$) reduced sleep latency and prolonged sleep duration in experimental mice in a dose-dependent manner compared to the control (vehicle) group. Mice in the control group exhibited a sleep latency of 27.60 ± 2.15 min and a total sleep duration of 152.00 ± 6.71 min. In contrast, mice treated with a lower dose of helicide (5 mg/kg) showed a markedly shorter sleep latency of 14.60 ± 2.37 min and an extended sleep duration of 161.60 ± 4.36 min. Increasing the dose to 10 mg/kg further reduced sleep latency to 9.40 ± 1.71 min while significantly enhancing total sleep duration to 174.80 ± 10.47 min. These results highlight the dose-dependent sedative effects of helicide. Meanwhile, the group treated with diazepam, exhibited significantly shorter sleep latency (6.20 ± 0.49 min) compared to the other groups, though slightly higher than the combination group (HEL-10 + diazepam, 5.20 ± 0.58 min). The total sleep duration in the diazepam group (183.20 ± 5.59 min) was also higher compared to other groups but slightly less (191.40 ± 3.86 min) than in the combination group (HEL-10 + diazepam).

In Silico Study

Molecular Docking

In silico studies showed that diazepam, the reference ligand, had the highest docking score at -8.7 kcal/mol with the GABA_A receptor. It formed one hydrogen bond with LEU285 and made hydrophobic bonds with the amino acid residues PHE289 (Pi-Pi Stacked), PRO233 (Alkyl), MET236 (Alkyl), LEU285 (Alkyl), PHE289 (Pi-Alkyl), MET286 (Pi-Alkyl), LEU232 (Pi-Alkyl), MET236 (Pi-Alkyl), and PRO233 (Pi-Alkyl). In contrast, the

experimental ligand HEL had a docking score of -7.7 kcal/mol, forming three hydrophobic bonds with ASN265, SER272, and THR266. Additionally, helicide interacted hydrophobically with residues PHE289 (Pi-Pi Stacked) and PRO233 (Pi-Alkyl). The details of docking scores, hydrophobic bonds, hydrogen bond length, and amino acid residues are summarized in Table 2, with Fig. 2 showing the 2D and 3D binding views.

Pharmacokinetics and Drug-likeness

The pharmacokinetics of helicide and diazepam were predicted *in silico* using the SwissADME and pkCSM web tool. Both compounds displayed favorable drug-likeness properties, qualifying them as potential drug candidates. Their molecular weights 284.26 g/mol for helicide and 284.74 g/mol for diazepam were within the acceptable range of less than 500 g/mol. The analysis also indicated that helicide is more water-soluble than diazepam. Furthermore, both helicide and diazepam demonstrated high gastrointestinal tract absorption. Additional pharmacokinetic parameters, including blood–brain barrier permeability, P-glycoprotein substrate status, and inhibition of cytochrome P450 2C19, are detailed in Table S1, with graphical representations provided in Fig. S1.

Toxicity

The ligand diazepam showed toxicity class 2 in the toxicology analysis, whereas the tested compound helicide was categorized under toxicity class 5. Additionally, diazepam showed an LD₅₀ of 48 mg/kg, while helicide (**1**) demonstrated a significantly higher LD₅₀ of 3750 mg/kg, indicating that this tested compound has a better safety profile. The toxicology study also demonstrated that both compounds exhibited positive (active) results in clinical toxicity. However, neither compound exhibited any toxic effects related

to hepatotoxicity, cardiotoxicity, immunotoxicity, mutagenicity, carcinogenicity, or nutritional toxicity, as they were inactive in all these areas. Various toxicity parameters of diazepam and helicide are listed in Table S2, and the network chart is presented in Fig. S1.

Central nervous system active drugs are commonly utilized in the therapeutic management of a range of neurological diseases and disorders (Lucas et al. 2006). Barbiturates, benzodiazepines, and ethanol are the primary agents used to depress the central nervous system. Both barbiturates and benzodiazepines exert their effects by interacting with postsynaptic GABA_A receptors (Ito et al. 1996; Khose et al. 2012). Barbiturates have a narrow safety margin, as a dose only 10 times higher than the therapeutic amount can be lethal in humans (Khan et al. 2014). Additionally, barbiturates are known to cause both physiological and psychological dependence (Essig 1964). In contrast, benzodiazepines can lead to tolerance and addiction. For example, diazepam, a widely used benzodiazepine, can cause dizziness in new users at doses between 5 and 10 mg. Long-term use of diazepam typically results in the development of tolerance (Mukty et al. 2024). Ethanol alters the flexibility of cell membranes and affects the GABAergic system (Burgos et al. 2015). It also leads to tolerance and physical dependence. As a result, finding safer and effective sedative drugs is essential.

The use of anxiolytic, sedative, or hypnotic drugs has been found to promote relaxation in animals by aiding in sleep (Nisar and Saleem 2023). The thiopental sodium-induced animal sleep model is used to explore the sedative and hypnotic effects related to GABA_A receptors. Thiopental sodium attaches to the barbiturate sites on GABA_A receptors, resulting in hyperpolarization of the postsynaptic membrane (Jakaria et al. 2017; Bappi et al. 2023). Benzodiazepines amplify the calming effect of thiopental sodium by acting as activators of GABA_A receptors (Campo-Soria et al. 2006). Substances that reduce the time it takes to fall asleep and extend sleep duration in thiopental sodium-induced rats may influence GABA_A receptors, particularly the $\alpha 1$ and $\beta 2$

Table 2 Molecular docking scores of helicide (**1**) and diazepam toward GABA_A (PDB) receptors

Ligands	Docking score (kcal/mol) with GABA _A (6X3X)	No. of Hydrogen bonds	Hydrogen bonds (HB)	Hydrogen bond length (Å)	Other bond residues
DZP	-8.7	1	LEU285	3.39	PHE289(Pi-Pi Stacked), PRO233(Alkyl), MET236(Alkyl), LEU285(Alkyl), PHE289 (Pi-Alkyl), MET286(Pi-Alkyl), LEU232(Pi-Alkyl), MET236(Pi-Alkyl), PRO233(Pi-Alkyl)
HEL	-7.7	3	ASN265, SER272, THR266,	3.21, 2.66, 3.02, 3.82	PHE289(Pi-Pi Stacked), PRO233(Pi-Alkyl)

DZP: diazepam, HEL: helicide

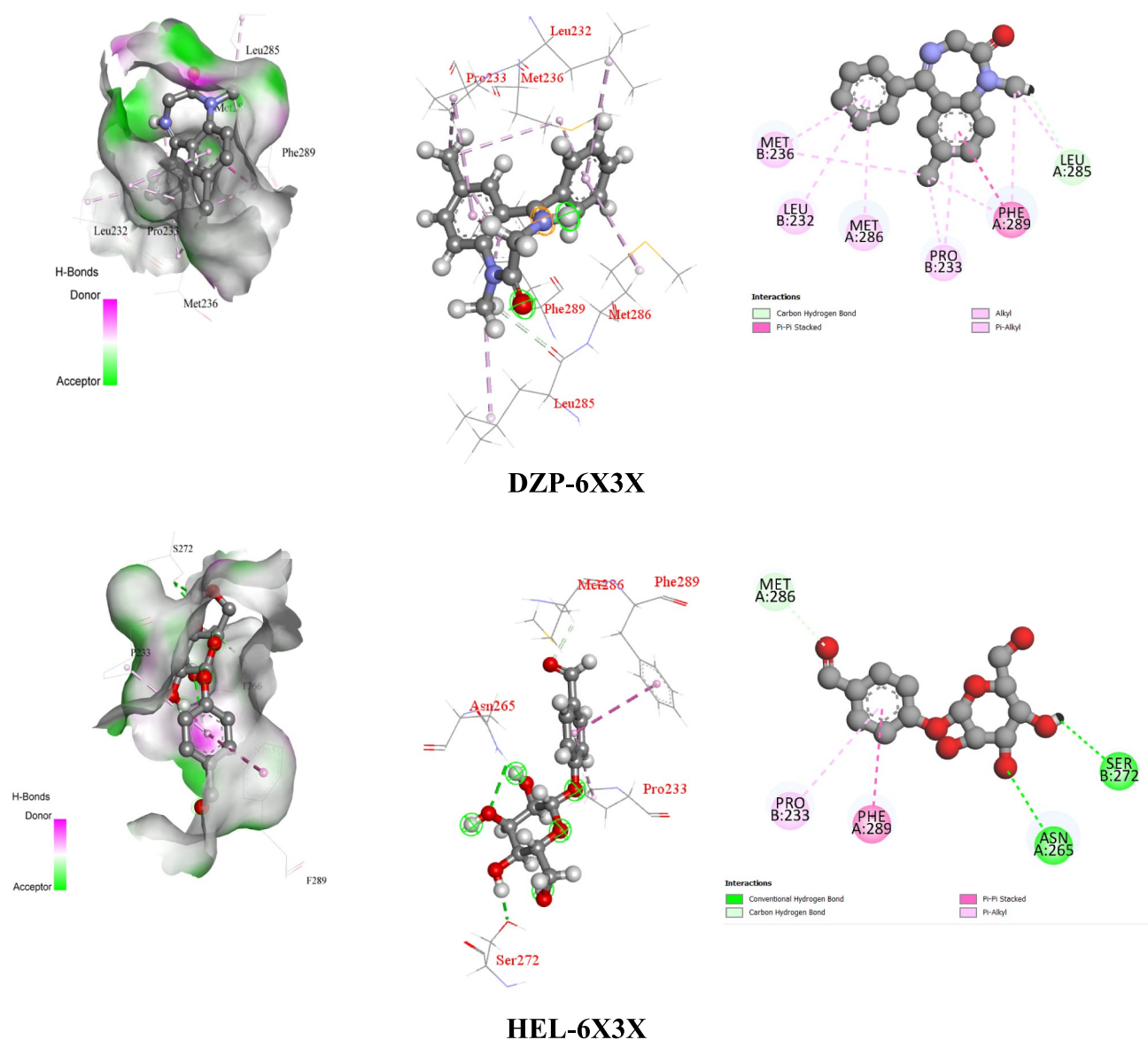


Fig. 2 2D and 3D visualization of diazepam and helicide interactions with the GABA_A receptor (PDB ID: 6X3X). Diazepam-6X3X (above) and helicide –6X3X (below)

subunits ((Raihan et al. 2011). While thiopental sodium was inducing sleep, helicide administration notably decreased the time needed to fall asleep and prolonged sleep duration. This suggests that helicide might have a GABAergic agonistic effect in animals.

Diazepam is commonly used as a reference drug to evaluate the anxiolytic effects of other substances. However, current evidence indicates that diazepam primarily functions as a sedative rather than an anxiolytic. Its sedative effects can be observed in rats at a dose of 2 mg/kg (*i.p.*) (Pádua-Reis et al. 2021). Studies have demonstrated that it enhances the activation of GABA_A receptors, which increases the entry of chloride ions into the cell (Qian et al. 2023). Hence,

substances that reduce sleep onset latency and extend sleep duration in diazepam-treated animals are likely sedatives that interact with GABA_A receptors (Bateson 2006; Pádua-Reis et al. 2021).

The research demonstrated that helicide reduced the sleeping time (9.40 ± 1.71 min) and increased total sleep duration (174.8 ± 10.47 min) in a dose-dependent manner in mice, suggesting interaction with GABA_A receptors. Computational results confirmed that helicide has a strong affinity (-7.7 kcal/mol) for the GABA_A receptor (PDB: 6X3X), particularly binding to the $\alpha 1$ and $\beta 2$ subunits. Synergistic effects occur when two or more substances produce a combined impact that exceeds the sum of their individual effects

(Ialongo et al. 2023). In our *in vivo* experiments, the combination of diazepam and helicide (10 mg/kg) significantly enhanced sleep duration (191.40 ± 3.86 min) and shortened sleep latency (5.20 ± 0.58 min) in mice, outperforming the effects of each substance when used alone. These results suggest a moderate sedative activity of helicide.

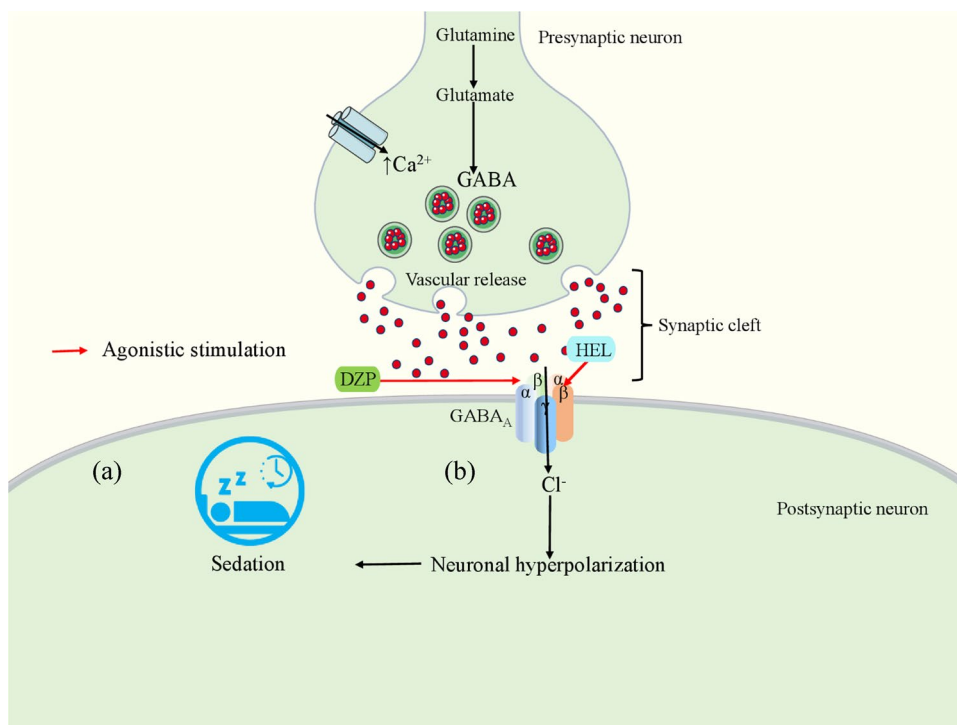
Molecular docking aims to predict the best way a drug fits with its target receptor, forming a stable complex when it binds together (Meng et al. 2011). Recent computational studies have revolutionized the way pharmaceutical candidates are designed, screened, and developed (Lin et al. 2020). This approach reduces the costs related to animal testing and laboratory work, while also shortening the overall evaluation time. Molecular affinity measures the strength of the interaction between a ligand and a specific protein (Niazi and Mariam 2023). In our molecular docking study, the tested ligand helicide displayed a strong binding affinity (-7.7 kcal/mol) to the GABA_A receptor (6X3X), which contains the $\alpha 1$ and $\beta 2$ subunits. The interaction resulted in the formation of three HBs, involving specific amino acid residues: ASN265, SER272, and THR266. Complex formation between biomacromolecules and their target ligands depends on hydrogen bonding, which plays a key role in determining binding selectivity (Duff Jr and Howell 2015). Additionally, hydrogen bonding enhances ligand binding and increases the therapeutic potential of these ligands (Chen et al. 2016). This molecular docking study revealed that helicide formed more hydrogen bonds compared to both a conventional drug and an agonist drug (diazepam).

The possible mechanisms of diazepam and helicide are illustrated in Fig. 3.

Drug-likeness evaluates potential drug candidates by analyzing physicochemical properties and structural features common to successful drugs, helping to identify compounds with undesirable ADMET profiles that may affect their effectiveness and safety (Tian et al. 2015; Agoni et al. 2020). It assesses how a drug behaves in the body by examining its physicochemical properties (Vistoli et al. 2008). Lipinski's rule of five is widely used for predicting pharmacokinetics and drug-likeness. Lipinski's rule of five states that a drug candidate should have a molecular weight of 500 g/mol or less, the numbers of hydrogen bond acceptor (HBA) and hydrogen bond donor (HBD) of helicide and diazepam within the limits ($HBA \leq 10$, $HBD \leq 5$), and lipophilicity (Log $P_{o/w}$) of no more than five (Chen et al. 2020). In addition, both ligands (helicide and diazepam) have molar refractivity (MR) within the range ($MR \leq 140$) of 65.97 and 87.95, respectively. According to Lipinski's rule of five, ligands should show favorable pharmacokinetic properties to be considered potential drug candidates. Our test ligand (helicide) meets all these criteria, demonstrating strong pharmacokinetic characteristics.

Toxicology testing is crucial for evaluating the potential adverse effects of substances. For instance, prolonged exposure to chemicals in humans can result in genetic damage, immune system impairment, cancer, and negative impacts on development and reproduction (Thompson et al. 2015; Sabarwal et al. 2018; Schuijt et al. 2021). The results of

Fig. 3 The suggested potential sedative mechanism of test compound helicide compared to the selected standard drugs diazepam. This figure illustrates the possible sedative mechanism of helicide and diazepam based on their binding affinity with $\alpha 1$ and $\beta 2$ subunits of the GABA_A receptor. Here, helicide and diazepam act as GABA receptor agonists. Abbreviations: HEL: helicide; DZP: diazepam; GABA: gamma aminobutyric acid



this toxicological study showed that helicide exhibited no toxic effects related to hepatotoxicity, neurotoxicity, carcinogenicity, respiratory toxicity, cardiotoxicity, immunotoxicity, nutritional toxicity, mutagenicity, cytotoxicity, or ecotoxicity. In addition, helicide exhibits adverse effects in terms of nephrotoxicity, and clinical toxicity *in silico*. When comparing the LD₅₀ of the test ligand (helicide) to that of diazepam, the test ligand shows a higher LD₅₀, meaning it is less toxic than helicide when given in the same amounts.

The primary limitation of our study is the potential influence of external factors such as light intensity and unexpected noise, despite our efforts to control these variables for each mice. Additionally, some mice may not exhibit stress responses to isolation or environmental changes due to inherent biological or psychological differences. As this is a comparative study, the findings do not conclusively establish the specific sedation mechanism of helicide. Rather, the proposed mechanism is based on computational data and observed sedative effects. Further research is needed to clarify the mechanisms of action, establish safety and efficacy, and determine optimal dosing before considering helicide for clinical use.

Conclusion

This study evaluates the sedative effects of helicide in a mice model subjected to thiopental sodium-induced sleep. The data demonstrate that helicide significantly enhances sleep duration and decreases sleep latency in a dose-dependent manner compared to the control group. Complementary *in silico* analyses support the drug-likeness of helicide, revealing its favorable pharmacokinetic properties. Molecular docking studies further elucidate that helicide exhibits a notable binding affinity for GABA_A receptors, with a binding energy of -7.7 kcal/mol. In comparison, the reference drug diazepam shows a slightly higher binding affinity with a binding energy of -8.7 kcal/mol. Importantly, the combination of helicide with diazepam results in a synergistic effect, leading to a more pronounced increase in sleep duration and a reduction in sleep latency compared to the administration of either compound alone. These findings underscore the potential of helicide as a candidate for further investigation. However, additional research is essential to clarify the precise mechanisms of action and to develop a comprehensive safety and efficacy profile, including the determination of optimal dosing strategies, prior to considering helicide for clinical use.

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Author Contributions MMK and MSB: conceptualization, methodology, software, and formal analysis; MMK, MSB and MMR: investigation; MMK and MMR: resources; MMK, MSB and MMR: writing—original draft preparation; MMK and MSB: mechanism drawing; MMK, MSB, MAI, SAA and HK: writing—review and editing; MTI: supervision; MMK, MSB and MTI: project administration. All authors have read and agreed to the published version of the manuscript.

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Data Availability Data are available from authors on request.

Declarations

Ethical Approval Ethical approval was obtained from the Department of Pharmacy and the Ethical Committee of Bangabandhu Sheikh Mujibur Rahman Science and Technology University (#bsmrstu-phr-19PHR016-02).

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